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Topological Space

1.1 Basic definition

Definition 1.1.1

Let X be a non-empty set. A set $\mathcal{T}$ of subsets of X is said to be a **topology** on X if

- (i) X and the empty set, $\emptyset$, belong to $\mathcal{T}$,
 - (ii) the union of any (finite or infinite) number of sets in $\mathcal{T}$ belongs to $\mathcal{T}$, and
 - (iii) the intersection of any two sets in $\mathcal{T}$ belongs to $\mathcal{T}$.
- The pair $(X, \mathcal{T})$ is called a **topological space**.

Definition 1.1.2

Let $(X, \mathcal{T})$ be any topological space. Then the members of $\mathcal{T}$ are said to be **open sets**.

Proposition 1.1.3

If $(X, \mathcal{T})$ is any topological space, then

- (i) X and $\emptyset$ are open sets,
- (ii) the union of any (finite or infinite) number of open sets is an open set,
- (iii) the intersection of any finite number of open sets is an open set.

Definition 1.1.4

Let $(X, \mathcal{T})$ be a topological space. A subset S of X is said to be a **closed set** in $(X, \mathcal{T})$ if its complement in X , namely $X \setminus S$, is open in $(X, \mathcal{T})$.

Proposition 1.1.5

If $(X, \mathcal{T})$ is any topological space, then

- (i) $\emptyset$ and X are closed sets,
- (ii) the intersection of any (finite or infinite) number of closed sets is a

closed set and
 (iii) the union of any finite number of closed sets is a closed set.

Definition 1.1.6

A subset $\mathcal{S}$ of a topological space $(X, \mathcal{T})$ is said to be **clopen** if it is both open and closed in $(X, \mathcal{T})$.

1.2 Basis for a Topology

Definition 1.2.1

Let $(X, \mathcal{T})$ be a topological space. A collection $\mathcal{B}$ of open subsets of X is said to be a **basis** for the topology $\mathcal{T}$ if every open set is a union of members of $\mathcal{B}$.

Remark.

拓扑基不是唯一的, 根据定义, 一旦一个集族 $\mathcal{B}$ 成为了一个拓扑基, 我们无论往 $\mathcal{B}$ 添加多少开集, 它还是这个拓扑的基. 因此直觉上, 拓扑的基是越小越好用的.

Note.

也就是说, 拓扑基是通过任意并运算生成拓扑的对象.

Proposition 1.2.2

Let X be a non-empty set and let $\mathcal{B}$ be a collection of subsets of X . Then $\mathcal{B}$ is a basis for a topology on X if and only if $\mathcal{B}$ has the following properties:

- (a) $X = \bigcup_{B \in \mathcal{B}} B$, and
- (b) for any $B_1, B_2 \in \mathcal{B}$, the set $B_1 \cap B_2$ is a union of members of $\mathcal{B}$.

Remark.

这里不谈与 X 上的任何一个具体的拓扑的关系, 只谈一个集族什么时候能够生成一个拓扑.

Note.

这个命题刻画了一个集族距离成为一组基有多远. 即一个集族需要什么额外条件, 才能通过任意并这个运算生成出一个拓扑. 命题中的条件 (a) 无非对应于 Definition 1.1.1 (i). 而通过基任意并生成的这个操作, 天然就是为了 Definition 1.1.1 (ii) 而建立的. 因此 (b) 就是为了满足 1.1.1 (iii) 而提出的条件. 我们发现生成拓扑对于有限交封闭的要求, 完全地变成了对于基的有限交的封闭的要求.

Proof. If $\mathcal{B}$ is a basis for a topology on X , say $\mathcal{T}_{\mathcal{B}}$. Then $X \in \mathcal{T}$ means that there is a collection of sets in $\mathcal{B}$, say $\mathcal{B}_1$, such that

$$X = \bigcup_{B \in \mathcal{B}_1} B \subseteq \bigcup_{B \in \mathcal{B}} B$$

And since every sets in $\mathcal{B}$ is a subset of X , it follows that (a) holds. To show (b), we only need to see that $\mathcal{B} \subseteq \mathcal{T}_{\mathcal{B}}$. Since $\mathcal{T}$ is closed under finite intersection, we have $B_1 \cap B_2 \in \mathcal{T}_{\mathcal{B}}$.

Then conversely, if $\mathcal{B}$ has the properties (a), (b), we only show that the collection $\langle \mathcal{B} \rangle$ of union of the members in $\mathcal{B}$, satisfies Definition 1.1.1 (iii). We take $A_1, A_2 \in \langle \mathcal{B} \rangle$, and suppose that

$$A_i = \bigcup_{j \in I_i} B_j, \quad B_j \in \mathcal{B}, \quad I_i \text{ is a index set, } i = 1, 2$$

Then

$$\begin{aligned} A_1 \cap A_2 &= \left(\bigcup_{j_1 \in I_1} B_{j_1} \right) \cap \left(\bigcup_{j_2 \in I_2} B_{j_2} \right) \\ &= \bigcup_{j_2 \in I_2} \left(\left(\bigcup_{j_1 \in I_1} B_{j_1} \right) \cap B_{j_2} \right) \\ &= \bigcup_{j_2 \in I_2} \left(\bigcup_{j_1 \in I_1} (B_{j_1} \cap B_{j_2}) \right) \\ &= \bigcup_{j_1 \in I_1, j_2 \in I_2} B_{j_1} \cap B_{j_2} \end{aligned}$$

Since every $B_{j_1} \cap B_{j_2}$ is a union of members of $\mathcal{B}$, and $A_1 \cap A_2$ is a union of such $B_{j_1} \cap B_{j_2}$, then $A_1 \cap A_2$ is a union of members of $\mathcal{B}$. $\square$

Proposition 1.2.3

Let $(X, \mathcal{T})$ be a topological space. A family $\mathcal{B}$ of open subsets of X is a basis for $\mathcal{T}$ if and only if for any point x belonging to any open set U , there is a $B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Remark.

表述 “there is a $B \in \mathcal{B}$ such that $x \in B \subseteq U$ ” 就是在说 “ U 可以由 $\mathcal{B}$ 通过 ‘并’ 生成” 的微观讲法. 带入这句话之下, 这个命题几乎就是一句 “废话”. 同样的道理也出现在 Proposition 1.2.4

Note.

想要看 $\mathcal{B}$ 是否作为基生成了拓扑 $\mathcal{T}$. 我们不用谈 $\mathcal{B}$ 中的元素如何如何 “并” 出每一个开集 U . 而是只需要 $\mathcal{B}$ 对于开集是足够 “精细” 的, 以至于对于任意的开集 U , 我们可以用 $\mathcal{B}$ 里面的元素在 U 上 “框出” 它的任意一个点.

Proof. If $\mathcal{B}$ is a basis for $\mathcal{T}$. Then we can write U as

$$U = \bigcup_{i \in I} B_i$$

Then for any $x \in U$, there exists $i_0 \in I$ such that $x \in B_{i_0} \subseteq U$.

Now we show the converse. First we show that $\mathcal{B}$ is a basis. We use Proposition 1.2.2 to verify this. Since X is an open set, for any $x \in X$, there is a $B_x \in \mathcal{B}$, such that $x \in B_x \subseteq X$, we get

$$X = \bigcup_{x \in X} B_x \subseteq \bigcup_{B \in \mathcal{B}} B \subseteq X$$

Thus

$$X = \bigcup_{B \in \mathcal{B}} B$$

Take any $B_1, B_2 \in \mathcal{B}$, we know $B_1 \cap B_2$ is again an open set. Then similarly, we get

$$B_1 \cap B_2 = \bigcup_{x \in B_1 \cap B_2} B_x$$

is a union of members in $\mathcal{B}$. Thus $\mathcal{B}$ is actually a basis. Finally, we only need to show that $\mathcal{B}$ generates $\mathcal{T}$, that is $\langle \mathcal{B} \rangle = \mathcal{T}$. $\langle \mathcal{B} \rangle$ is the collection of unions

of members in $\mathcal{B}$. $\langle \mathcal{B} \rangle \subseteq \mathcal{T}$ is obvious. $\mathcal{T} \subseteq \langle \mathcal{B} \rangle$ concludes from

$$U = \bigcup_{x \in U} B_x \in \langle \mathcal{B} \rangle$$

□

Proposition 1.2.4

Let $\mathcal{B}$ be a basis for a topology $\mathcal{T}$ on a set X . Then a subset U of X is open if and only if for each $x \in U$ there exists a $B \in \mathcal{B}$ such that $x \in B \subseteq U$.

Note.

把基 $\mathcal{B}$ 看成是 $\mathcal{T}$ 中代表着“足够精细”的某一批. 验证 U 是否是开集, 只需要看 U 是否可以被这一小批集合“概括”.

Proof. $\implies$ is immediately from Proposition 1.2.3. Now suppose that for each $x \in U$, there exists a $B_x \in \mathcal{B}$ such that $x \in B_x \subseteq U$, then

$$U = \bigcup_{x \in U} B_x \in \langle \mathcal{B} \rangle = \mathcal{T}$$

□

Proposition 1.2.5

Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be bases for topologies $\mathcal{T}_1$ and $\mathcal{T}_2$, respectively, on a non-empty set X . Then $\mathcal{T}_1 = \mathcal{T}_2$ if and only if

1. for each $B \in \mathcal{B}_1$ and each $x \in B$, there exists a $B' \in \mathcal{B}_2$ such that $x \in B' \subseteq B$, and
2. for each $B \in \mathcal{B}_2$ and each $x \in B$, there exists a $B' \in \mathcal{B}_1$ such that $x \in B' \subseteq B$.

Proof sketch.

考虑这句话: 表述 “there is a $B \in \mathcal{B}$ such that $x \in B \subseteq U$ ” 就是在说 “ U 可以由 $\mathcal{B}$ 通过 ‘并’ 生成”. 1. 是说 $\mathcal{B}_1$ 中任意集合可以由 $\mathcal{B}_2$ “并” 出来. 自然地 $\mathcal{T}_1$ 中任意集合也被 $\mathcal{B}_2$ “并出来”. 因此 1. 就是 $\mathcal{T}_1 \subseteq \mathcal{T}_2$. 同样地, 2. 就是 $\mathcal{T}_2 \subseteq \mathcal{T}_1$.

Lemma 1.2.6

Let $\mathcal{B}$ and $\mathcal{B}'$ be bases for the topologies $\mathcal{T}$ and $\mathcal{T}'$, respectively, on X . Then the following are equivalent:

1. $\mathcal{T}'$ is finer than $\mathcal{T}$.
2. For each $x \in X$ and each basis element $B \in \mathcal{B}$ containing x , there is a basis element $B' \in \mathcal{B}'$ such that $x \in B' \subset B$.

Proof sketch.

同样的, 2. 就是在说每个 B 可以由一堆 B' 并出来. 那么显然每个 $U \in \mathcal{T}$ 可以由一堆 B' 并出来, 即 $U' \in \mathcal{T}'$.

1.3 Closed Sets and Limit Points

Definition 1.3.1

A subset A of a topological space X is said to be **closed** if the set $X - A$ is open.

1.3.1 Closure, Interior

Definition 1.3.2

Given a subset A of a topological space X

- The **interior** of A , denoted by A° or by $\text{Int } A$, is defined as the union of all open sets contained in A .
- The **closure** of A , denoted by $\bar{A}$ or by $\text{Cl } A$, is defined as the intersection of all closed sets containing A .

Theorem 1.3.3

Let A be a subset of the topological space X .

- (a) Then $x \in \bar{A}$ if and only if every open set U containing x intersects A .
- (b) Supposing the topology of X is given by a basis, then $x \in \bar{A}$ if and only if every basis element B containing x intersects A .

Proof. (a) Method 1. Note that

$$V \text{ is a closed set containing } A \iff V^c \text{ is an open set such that } V^c \cap A = \emptyset$$

We have

$$\begin{aligned}
 x \notin \bar{A} &\iff x \notin \bigcap_{V \text{ is a closed set containing } A} V \\
 &\iff x \in \bigcup_{V \text{ is a closed set containing } A} V^c \\
 &\iff x \in \bigcup_{U \text{ is an open set such that } U \cap A = \emptyset} U \\
 &\iff \text{there exists open set } U \text{ such that } x \in U, U \cap A = \emptyset
 \end{aligned}$$

Method 2. Suppose that $x \in \bar{A}$, then x is contained in any closed set containing A . If there exists open set U containing x does not intersect A . Then U^c is a closed set containing A , which implies that $x \in U^c$, a contradiction. Conversely, if every open set containing x intersects A , then every closed set that x does not lie in cannot contain A . i.e. x lies in every closed set containing A .

- (b) If every open set containing x intersects A , then every basis B containing x intersects A since B is also an open set. Conversely, if every basis B containing x intersects A , since every open set U contains a basis containing x , it intersects with A as well. Then it follows from (a).

□

1.3.2 Limit Points

Definition 1.3.4

If A is a subset of the topological space X and if x is a point of X , we say that x is a **limit point** (or “cluster point,” or “point of accumulation”) of A if every neighborhood of x intersects A in some point **other than x itself**. Said differently, x is a limit point of A if it belongs to the closure of $A - \{x\}$.

Theorem 1.3.5

Let A be a subset of the topological space X ; let A' be the set of all **limit points** of A . Then

$$\bar{A} = A \cup A'.$$

Proof sketch.

对于 $\bar{A} \subseteq A \cup A'$, $A \subseteq \bar{A}$ 显然而 $A' \subseteq \bar{A}$ 是 Theorem ?? 的直接推论. $A \cup A' \subseteq \bar{A}$, 则考虑若 x 不在 A 里面, 那么 $x \in A'$ 的定义中的开集一定与 A 相交, 从而有 $x \in \bar{A}$.

Continuous Functions

Continuous Functions

Homeomorphisms

3.1 Subspaces

Definition 3.1.1

Let Y be a non-empty subset of a topological space $(X, \mathcal{T})$. The collection $\mathcal{T}_Y = \{O \cap Y : O \in \mathcal{T}\}$ of subsets of Y is a topology on Y called the *subspace topology* (or the *relative topology* or the *induced topology* or the *topology induced on Y by $\mathcal{T}$*). The topological space $(Y, \mathcal{T}_Y)$ is said to be a *subspace* of $(X, \mathcal{T})$.

Limit Points

4.1 Limit Points and Closure

Definition 4.1.1

Let A be a subset of a topological space $(X, \mathcal{T})$. A point $x \in X$ is said to be a **limit point** (or **accumulation point** or **cluster point**) of A if every open set, U , containing x contains a point of A different from x .

Note.

A 的极限点 x 是不能与 $A \setminus \{x\}$ “分离开”的点. 也就是说在每一个开集里面, x 和 $A \setminus \{x\}$ 要么都有在里面的点, 要么都没有.

Proposition 4.1.2

Let A be a subset of a topological space $(X, \mathcal{T})$. Then A is closed in $(X, \mathcal{T})$ if and only if A contains all of its limit points.

Proof sketch.

如果集合 A 是闭的, 那么 A 外面的点 x 都被落在 A 外面的一个开集 U 给包住了, 那么就导致了 x 不是极限点.

如果 A 包含了所有极限点, 那么 A 外面的点 x 都不是极限点, 从而 x 一定能被与 A 无交的一个开集给包住.

Proof. If A contains all of its limit points. Take any $x \in A^c$, then x is not a limit point of A . There exists open set U , containing x contains no point of A . Then $U \subseteq A^c$, which implies that A^c is an open set. Thus A is a closed set.

Now, let A be a closed set. Let x be any limit point of A . If $x \notin A$, then there is a open set $U \subseteq A^c$ such that $x \in U$. By the definition of the limit point, there is a point y of A different from x belongs to U , which is a contradiction to $U \cap A = \emptyset$. $\square$

Proposition 4.1.3

Let A be a subset of a topological space $(X, \mathcal{T})$ and A' the set of all limit points of A . Then $A \cup A'$ is a closed set.

Remark.

The proof shows that : If U is an open set, and A an arbitrary set. Then $U \cap A = \emptyset$ implies that $U \cap A' = \emptyset$.

Proof. Let $p \in X \setminus (A \cup A')$, then p neither a point in A nor a limit point of A . Thus there exists a open set U , such that $p \in U$, $U \cap A = \emptyset$. We now claim that $U \cap A' = \emptyset$, otherwise, we take $q \in U \cap A'$. Then U contains a point of A different from q , which is a contradiction to $U \cap A = \emptyset$. Thus we have

$$U \cap (A \cup A') = \emptyset \implies p \in U \subseteq X \setminus (A \cup A')$$

Then $X \setminus (A \cup A')$ is an open set. □

Definition 4.1.4

Let A be a subset of a topological space $(X, \mathcal{T})$. Then the set $A \cup A'$ consisting of A and all its limit points is called the **closure of A** and is denoted by $\bar{A}$.

Definition 4.1.5

Let A be a subset of a topological space $(X, \mathcal{T})$. Then A is said to be **dense** in X or **everywhere dense** in X if $\bar{A} = X$.

Proposition 4.1.6

Let A be a subset of a topological space $(X, \mathcal{T})$. Then A is dense in X if and only if every non-empty open subset of X intersects A non-trivially (that is, if $U \in \mathcal{T}$ and $U \neq \emptyset$ then $A \cap U \neq \emptyset$.)

Proof. If A is dense in X , then $\bar{A} = X$. Take any non-empty open subset U of X . If $A \cap U = \emptyset$, then $A' \cap U = \emptyset$ from Proposition 3.1.3 Remark . Thus $U \cap \bar{A} = \emptyset$, then $U \cap X = \emptyset$, which is a contradiction. Thus $A \cap U \neq \emptyset$.

Conversely, if for any $U \in \mathcal{T}$, and $U \neq \emptyset$, we have $A \cap U \neq \emptyset$. Take any $x \in X$. If $x \notin A$, then for any open set $V \in \mathcal{T}$, containing x , we have

$A \cap V \neq \emptyset$. And we have $V \cap (A \setminus \{x\}) \neq \emptyset$. Thus x is a limit point of A . It follows that $x \in A \cup A' = \bar{A}$, $X \subseteq \bar{A}$. Then $X = \bar{A}$. $\square$

Metric Spaces

5.1 Convergence of Sequences

Definition 5.1.1 ► Convergence

Let (X, d) be a metric space and $x_1, \dots, x_n, \dots$ a sequence of points in X . Then the sequence is said to **converge to** $x \in X$ if given any $\varepsilon > 0$ there exists an integer n_0 such that for all $n \geq n_0$, $d(x, x_n) < \varepsilon$. This is denoted by $x_n \rightarrow x$.

The sequence $y_1, y_2, \dots, y_n, \dots$ of points in (X, d) is said to be **convergent** if there exists a point $y \in X$ such that $y_n \rightarrow y$.

Theorem 5.1.2

Let (X, d) be a metric space. $A \subseteq X$ is a subset, $p \in A$. Then p is a limit point of A , if and only if there exists a sequence $\{a_n\}$ composed with distinct points in A converging to p .

Proof. If p is a limit point of A . Then $B_d(p, 1) \cap (A \setminus \{p\}) \neq \emptyset$, we choose a point a_1 there. If we have points $a_1, a_2, \dots, a_n$ such that $a_i \in B_d\left(p, \frac{1}{i}\right) \cap B_d(p, m_i)$, where

$$m_i := \min \{d(p, a_j)\}_{1 \leq j \leq i}$$

Since the sets $\left(B_d\left(p, \frac{1}{n+1}\right) \cap B_d(p, m_n)\right) \cap (A \setminus \{p\})$ is nonempty, we can take a point in it. We inductively give a sequence $\{a_n\}$ composed with distinct points in A .

If $\{a_n\}$ is such sequence. Let U be any open set containing p . Since $\{B_d(x, \varepsilon) : x \in X, \varepsilon > 0\}$ is a basis for (X, d) . We have from Proposition 1.2.4 that there exists a ball B such that $p \in B \subseteq U$. Then it is easy to verify that there exists $B_d(p, \varepsilon)$ such that $p \in B_d(p, \varepsilon) \subseteq B$. Since the sequence $\{a_n\}$ in A converges to p , there exists some (in fact, infinitely many) $a_n \in B_d(p, \varepsilon)$ that is not equal to p . i.e, $p \in B_d(p, \varepsilon) \cap (A \setminus \{p\}) \subseteq U \cap (A \setminus \{p\}) \neq \emptyset$. $\square$

Proposition 5.1.3

Let (X, d) be a metric space. A subset A of X is closed in (X, d) if and only if every convergent sequence of points in A converges to a point in A . (In other words, A is closed in (X, d) if and only if $a_n \rightarrow x$, where $x \in X$ and $a_n \in A$ for all n , implies $x \in A$.)

5.2 Completeness

Definition 5.2.1 ▶ Cauchy sequence

A sequence $x_1, x_2, \dots, x_n, \dots$ of points in a metric space (X, d) is said to be a **Cauchy sequence** if given any real number $\varepsilon > 0$, there exists a positive integer n_0 , such that for all integers $m \geq n_0$ and $n \geq n_0$, $d(x_m, x_n) < \varepsilon$.

Proposition 5.2.2

Let (X, d) be a metric space and $x_1, x_2, \dots, x_n, \dots$ a sequence of points in (X, d) . If there exists a point $a \in X$, such that the sequence converges to a , that is, $x_n \rightarrow a$, then the sequence is a Cauchy sequence.

Definition 5.2.3

A metric space (X, d) is said to be **complete** if every Cauchy sequence in (X, d) converges to a point in (X, d) .

Proposition 5.2.4

Let (X, d) be a metric space, Y a subset of X , and d_1 the metric induced on Y by d .

- (i) If (X, d) is a complete metric space and Y is a closed subspace of (X, d) , then (Y, d_1) is a complete metric space.
- (ii) If (Y, d_1) is a complete metric space, then Y is a closed subspace of (X, d) .

Countability

Definition 6.0.1 ► Second-Countable Space

topological space X is called second-countable if it has a countable basis for its topology.

Definition 6.0.2 ► Lindelöf Space

A topological space X is called a Lindelöf space if every open cover of X has a countable subcover.

Theorem 6.0.3 ► Lindeöf Theorem

Every second-countable space is Lindeöf.

Proof. Let X be a second-countable space, $\mathcal{B} = \{B_n\}_{n=1}^{\infty}$ is a countable basis of X . Let $\mathcal{U} = \{U_{\alpha}\}_{\alpha \in A}$ be any open cover of X . Let

$$\mathcal{B}' := \{B_n \in \mathcal{B} : \exists U_{\alpha} \in \mathcal{U}, \text{ s.t. } B_n \subseteq U_{\alpha}\}$$

For each $x \in X$, $\exists U_{\alpha} \in \mathcal{U}$ such that $x \in U_{\alpha}$. Since $\mathcal{B}$ is a basis, there $\exists B_n \in \mathcal{B}$ such that $x \in B_n \subseteq U_{\alpha}$. $B_n \in \mathcal{B}'$ by definition. Thus $\mathcal{B}'$ covers X . For each $B_{n_k} \in \mathcal{B}'$, we choose an open set U_k in $\mathcal{U}$, such that $B_{n_k} \subseteq U_k$. Then $\{U_k\}_{k=1}^{\infty}$ is a subcover of $\mathcal{U}$. $\square$

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